

Task 1

Operating system	Windows 11
CPU	AMD Ryzen 5 7533HS with Radeon Graphics
RAM	24GB
Programming language	Python 3.12.7
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Glob, Json

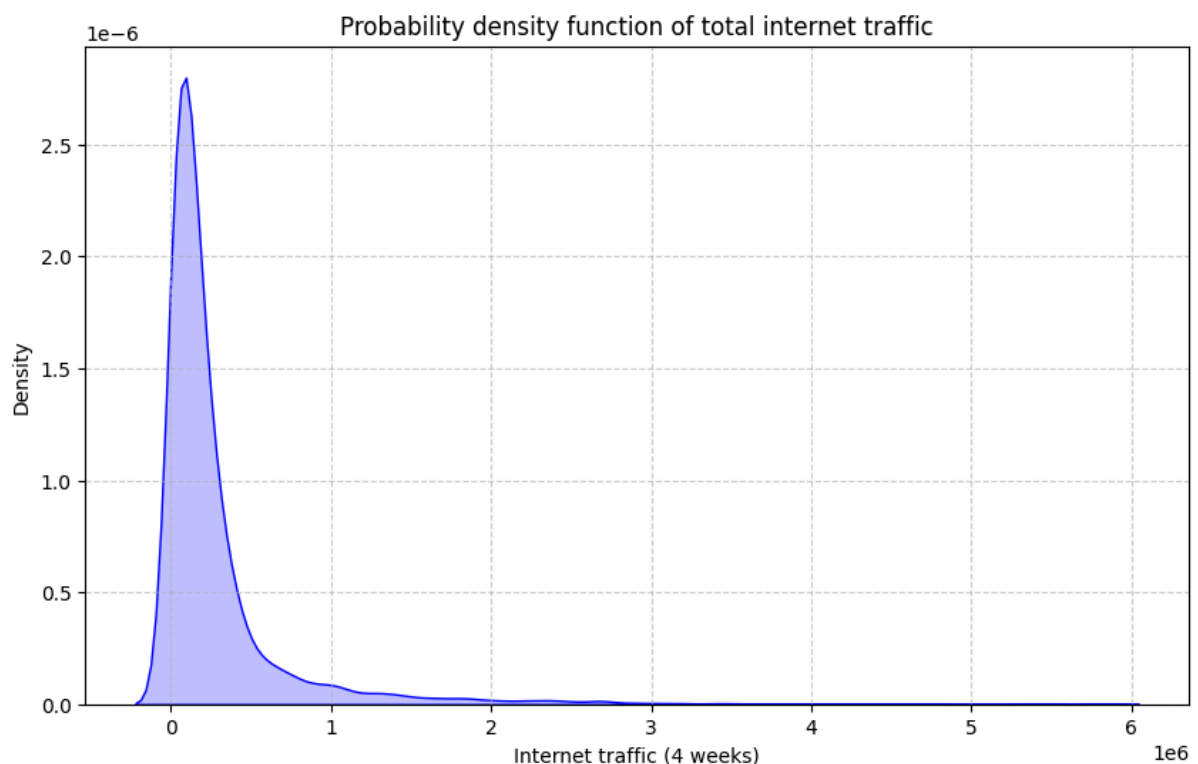
Task 2

In order to ensure high-quality data, the following steps were taken:

- The data was loaded such that “Country Code” was treated as the third field, which allowed the columns to be extracted correctly.
- The `dropna(subset=['Internet'])` function was used to remove records in which no internet activity was recorded.
- The “TimeInterval” column has been converted from a numeric format (milliseconds) to a readable date and time format (datetime) using the `pd.to_datetime` function with the unit set to 'ms'

The very large difference between the mean and the median, along with the high standard deviation, indicates a highly skewed distribution. This suggests that most areas generate very little traffic, while a small number of high-traffic areas account for the vast majority of online activity.

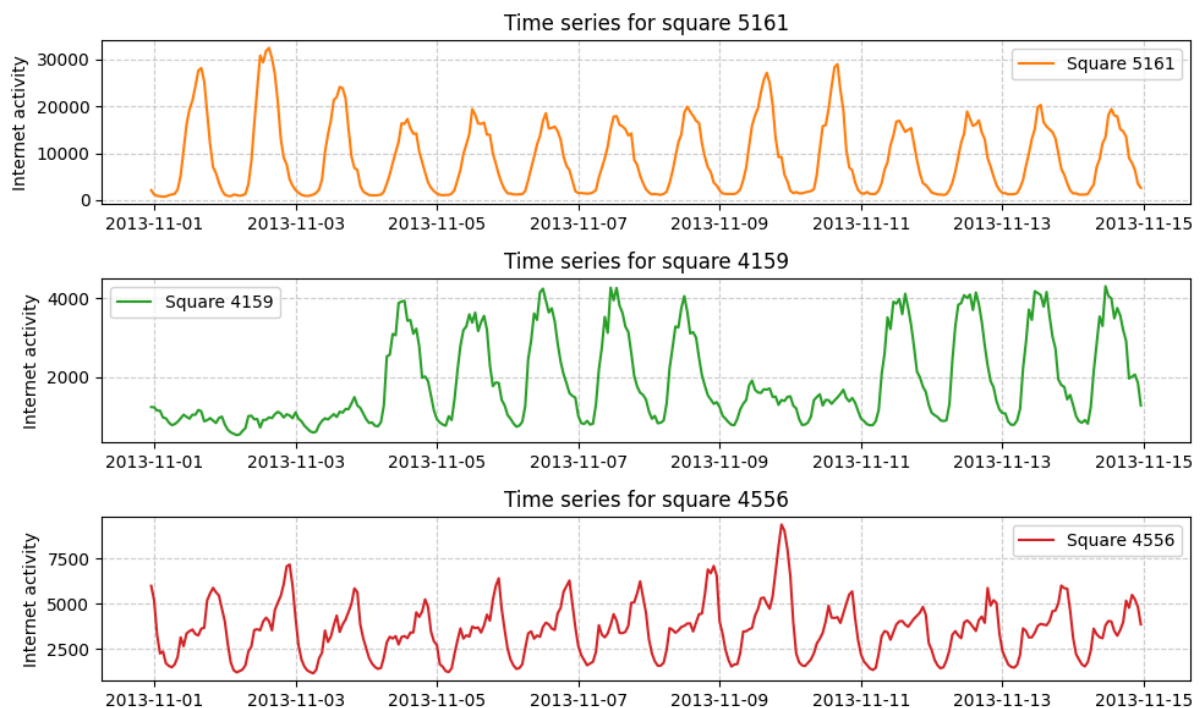
Task 3



Commentary on the results:

- The graph shows a highly heterogeneous and asymmetric distribution.
- Most of the 10,000 areas fall within the low-traffic range, while only a very small number of areas show extremely high activity.
- Areas with low traffic volume include sparsely populated zones, industrial outskirts, and parks. In contrast, areas with high traffic volume include densely populated residential neighborhoods, major business districts, and locations where mobile network usage is exceptionally high.

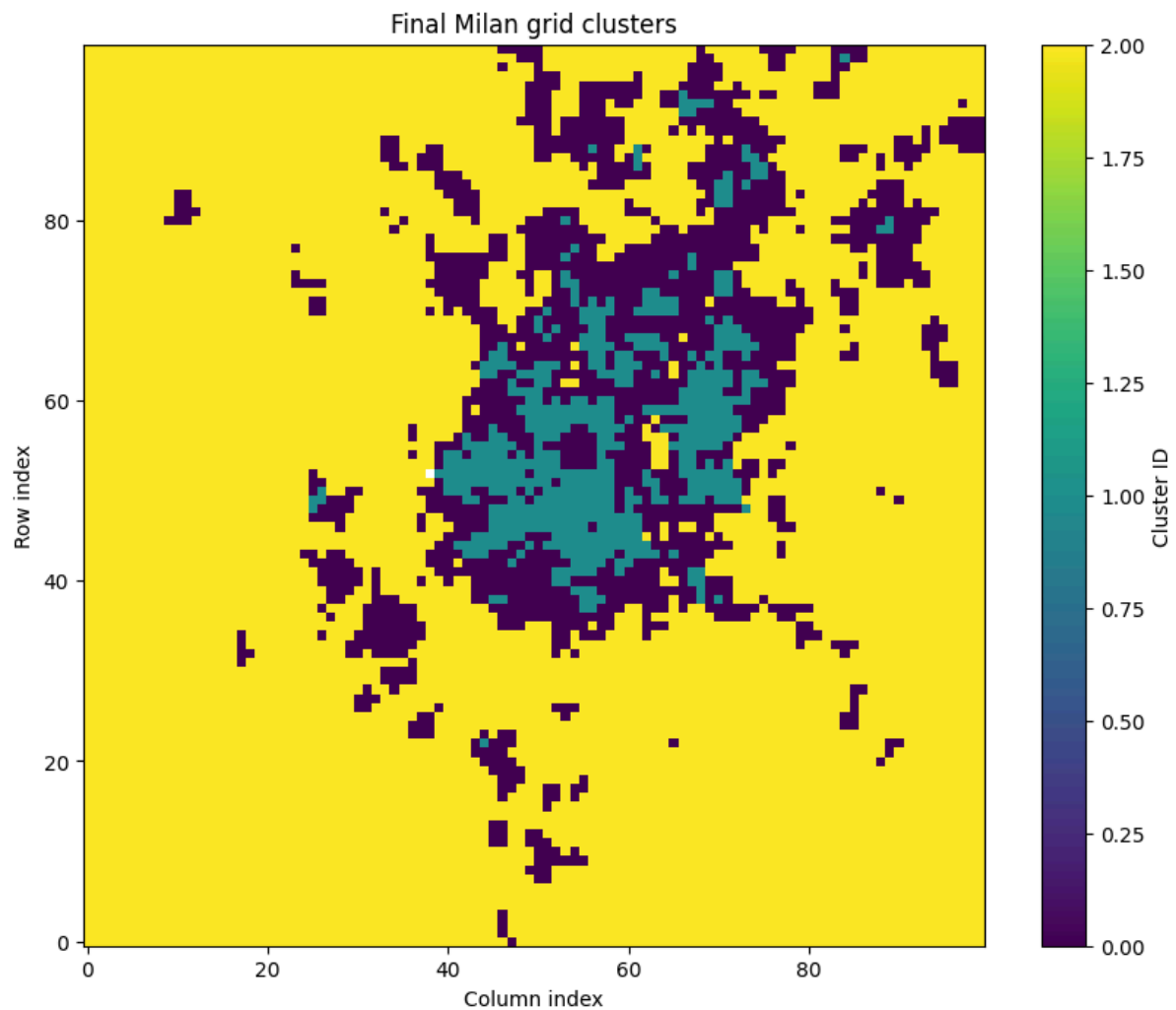
Task 4



Commentary on the results:

- Square 5161 generates the highest foot traffic, so it is likely the heart of the city or a major transportation hub.
- The chart for square 4556 is very “jagged”. The peaks aren't smooth, they have many smaller spikes. Traffic doesn't drop off as sharply in the afternoon. This is a typical pattern for a residential or entertainment district where people, for example, watch movies and use the internet after work.
- This chart for square 4159 shows the traffic pattern for an office and industrial area, where a distinct cycle of five high weekday peaks and two low weekend peaks (the “5+2” pattern) indicates that traffic is closely tied to working hours.

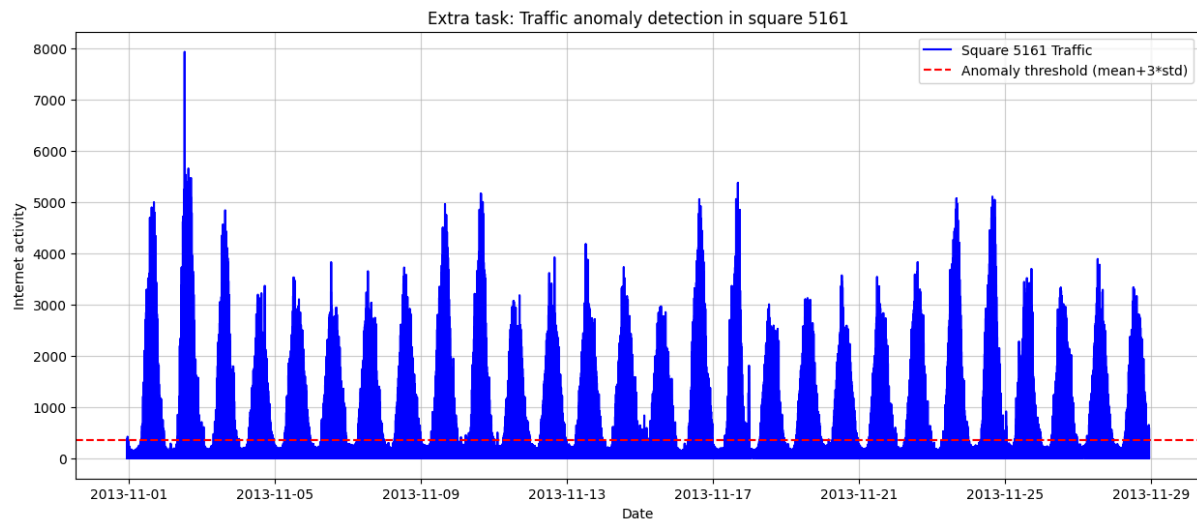
Task 5



Commentary on the results:

- The concentration of high-density clusters in the central part of the map confirms that Milan has a single strong business and service hub, from which areas of lower traffic density radiate outwards.
- The high territorial cohesion of the clusters (the absence of scattered, isolated points of a different color) demonstrates that people's digital behaviors are widespread and repetitive.

Extra task



Commentary on the results:

- The use of statistical methods identified grid square 5161 as the most anomalous among all 10,000 grid squares in Milan.
- The main reason for this classification is an extreme spike in activity in early November, which was more than double the standard daily peaks for this location.
- After the main anomaly occurred, activity returned to its regular daily cycle in the following days of November, suggesting that the detected event was isolated

For the programming tasks, I used an LLM model (Gemini 3 Flash), which helped me develop the logic needed to efficiently load a large number of text files and correctly interpret the structure of GeoJSON files, which I was working with for the first time. The model also assisted in performing statistical calculations (Z-score, developing a sorting procedure, and selecting the 10 areas with the highest number of anomalies). This tool facilitated the creation of code based on the glob, json, and pandas libraries, which enabled the efficient generation of technical results used for independent substantive analysis.